



PH 302: Introduction to Statistical Mechanics

Course Code	PH 302
Course Name	Introduction to Statistical Mechanics
L-T-P-C	2.5-0.5-0-3
Approval	2nd Adhoc Senate (May 2012), in the Minor in Applied Physics curriculum

Course Contents

- **Module I**
 - Statistical concepts and examples- random walk problem in one dimension- mean values- probability distribution for large N. Probability distribution many variables. [6 Lectures]
- **Module II**
 - Statistical description of a system of particles - Statistical ensemble- Microstate and macrostate - Density of states. Connection between statistics and thermodynamics - Relation between number of macrostates and entropy-classical ideal gas, Gibb's paradox. [6 Lectures]
- **Module III**
 - Liouville's theorem- Phase space and connection between mechanics and statistical mechanics Micro-canonical ensemble - Computational methods to calculate phase space trajectory- Molecular dynamics and Monte Carlo methods. [6 Lectures]
- **Module IV**
 - Canonical ensemble- partition function. Thermodynamics from the partition function- Helmholtz free energy. Classical Ideal gas- equipartition and virial theorem. System of harmonic oscillators and spin systems. Grand canonical ensemble- density and energy fluctuations- Gibbs free energy. [6 Lectures]
- **Module V**
 - Formulation of quantum statistical mechanics- density matrix- micro-canonical, canonical and grand canonical ensembles- Systems composed of indistinguishable particles, Slater determinant. [6 Lectures]
- **Module VI**
 - Maxwell-Boltzmann , Fermi-Dirac, and Bose-Einstein statistics - Ideal gas in classical and quantum. ensembles -Ideal Bose systems Black body radiation-lattice vibrations in solids- Ideal Fermi systems -magnetic systems- Pauli paramagnetism-Landau diamagnetism- electron gas in metals. [6 Lectures]
- **Module VII**
 - Brownian motion - Langevin equation - Fluctuation-dissipation theorem-correlation functions and friction coefficient. [4 Lectures]

References

1. F. Reif, **Fundamentals of statistical and thermal physics**
2. K. Huang, **Introduction to statistical physics**
3. F Mandl, **Statistical physics**
4. R K Pathria, **Statistical Mechanics**
5. K Huang, **Statistical Physics**